The documentation and process conversion measures necessary to comply with this revision shall be completed by 22 Oct 94.

METRIC

MIL-S-19500/610B
22 July 1994
SUPERSEDING
MIL-S-19500/610A
21 March 1994

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, HERMETIC, DIODE, SILICON, SCHOTTKY BARRIER,
TYPES 1N6677-1 AND 1N6677UR-1
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, Schottky barrier diodes. Four levels of product assurance are provided for each device type as specified in MIL-S-19500. Two levels of product assurance for die (element evaluation).

1.2 Physical dimensions. See figure 1 (DO-35), figure 2 (DO-213AA), and figure 3 (JANC die) dimensions.

1.3 Maximum ratings.

Types	V_{RWM}	I_{O1} 1/	I_{FSM}	T_{STG}	T_J
	V (pk)	mA dc	A (pk)	C°	C°
1N6677-1, 1N6677UR-1	40	200	5	-65 to +150	-65 to +125

1/ Derate linearly at 8 mA/°C above T_L or T_{EC} = +100°C where T_L is at L = 9.52 mm (.375 inch).

1.4 Primary electrical characteristics. Unless otherwise specified, primary electrical characteristics at T_A = +25°C.

Types	Max V_{FM1} I_{FM} = 20 mA	Max V_{FM2} I_{FM} = 200 mA	Max V_{FM3} I_{FM} = 630 mA	Max I_{RM} V_{RM} = 40 V dc pulsed method (see 4.5.1)		Max $R_{\theta JL}$ or $R_{\theta JEC}$ 9.52 mm (.375") lead length or end cap	Max $Z_{\theta JX}$	Max C_T V_R = 0 V dc
				T_J = +25°C I_{RM1}	T_J = +100°C I_{RM2}			
	V (pk)	V (pk)	V (pk)	µA	mA	°C/W	°C/W	pF
1N6677-1	0.37	0.50	0.70	5.0	0.60	250	20	50
1N6677UR-1	0.37	0.50	0.70	5.0	0.60	100	20	50

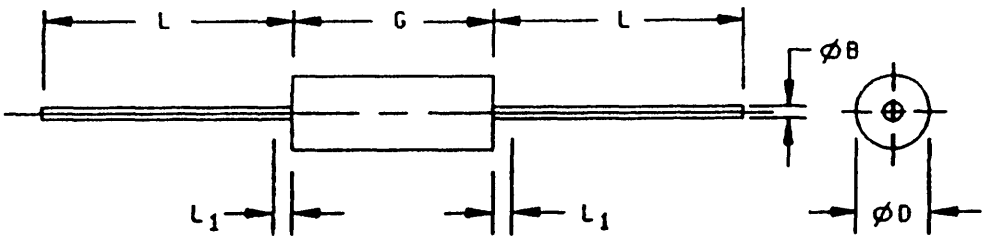
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELD, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:07Z
Check the source to verify that this is the current version before use.

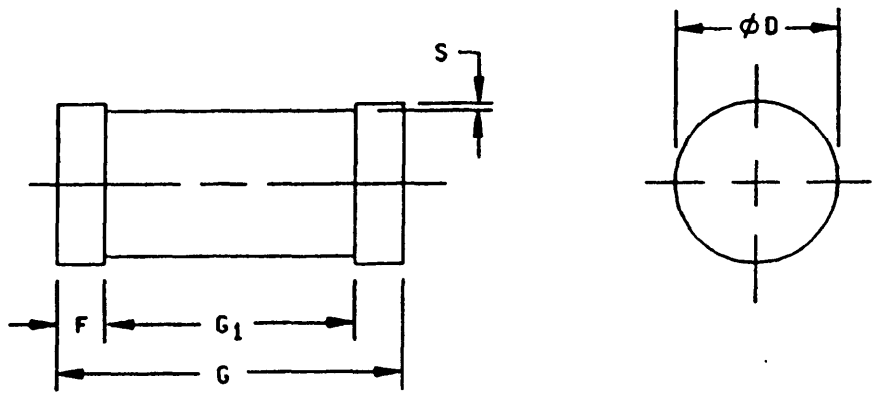


Symbol	Dimensions				Notes
	Millimeters		Inches		
	Min	Max	Min	Max	
ØB	0.46	0.56	.018	.022	
ØD	1.40	2.29	.055	.090	3
G	3.05	5.08	.120	.200	3
L	25.40	38.10	1.000	1.500	
L_1		1.27		.050	4

NOTES:
1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. Package contour optional within ØD and length G. Heat slugs, if any, shall be included within this cylinder but shall not be subject to minimum limit of ØD.
4. Within this zone lead, diameter may vary to allow for lead finishes and irregularities other than heat slugs.

FIGURE 1. Physical dimensions of 1N6677-1 (DO-35).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:07Z
Check the source to verify that this is the current version before use.

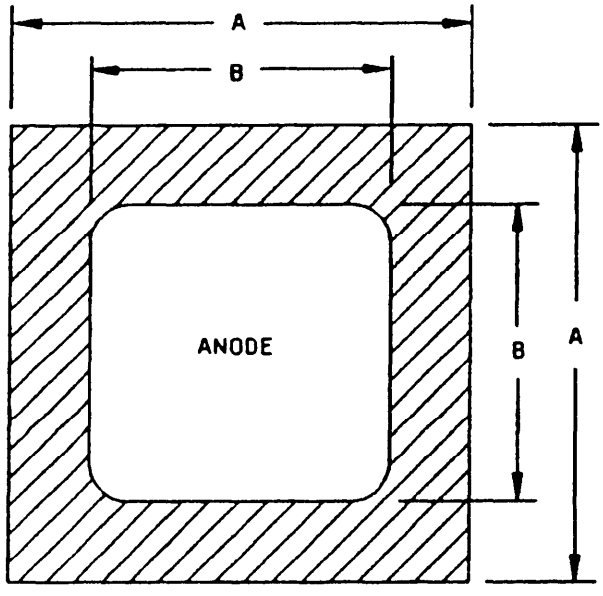


Symbol	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
ϕD	1.60	1.70	.063	.067
F	0.41	0.56	.016	.022
G	3.30	3.71	.130	.146
G_1	2.54 ref		.100 ref	
S	0.03 min		.001 min	

NOTES:
1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.

FIGURE 2. Physical dimensions of surface mount family, 1N6677UR-1 (DO-213AA).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:07Z
Check the source to verify that this is the current version before use.



BACKSIDE IS CATHODE

A version

Symbol	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
A	0.53	0.64	.021	.025
B	0.33	0.43	.013	.017

Design data

Metallization:
Top: (Anode) . . . . . AL
Back: (Cathode). . . . . Au

AL thickness . . . . . 25,000 Å minimum
Gold thickness . . . . . 4,000 Å minimum
Chip thickness . . . . . 0.254 mm (10 mils) ±0.051 mm (2 mils)

FIGURE 3. JANC (A-version) die dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:07Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500, and as follows:

JANH	High reliability product assurance level for unencapsulated devices.
JANK	Space reliability product assurance level for unencapsulated devices.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figures 1, 2, and 3 herein. Plastic packages are prohibited. The US Government preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-S-19500. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.3.2 Diode construction. All devices shall be metallurgically bonded, double plug construction in accordance with the requirements of category I or II (see MIL-S-19500).

3.3.2.1 Surface mount. The U version shall be considered structurally identical to the non-U version except for lead attach.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

3.4.1 Marking for UR devices. UR-suffix parts are to be marked with a cathode band only. Initial container package marking will be in accordance with MIL-S-19500.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:07Z
Check the source to verify that this is the current version before use.

as adversely affecting the qualified-product status of the device, or applicable JAN marking.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.2.1 Construction verification. Cross sectional photos from three devices shall be submitted in the qualification report.

4.2.2 JANHC and JANKC devices. Qualification for JANC devices shall be in accordance with appendix H of MIL-S-19500. This testing may be performed on a TO-5 package in lieu of the axial leaded package.

4.3 Screening (JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement		
	JANS level	JANTX and JANTXV levels	JAN level
3a	Temperature cycling	Temperature cycling	Temperature cycling in accordance with MIL-S-19500, JANTX level.
3c 1/	Thermal impedance (see 4.5.3)	Thermal impedance (see 4.5.3)	Thermal impedance (see 4.5.3)
9	I_{R1} and V_{FM2}	Not applicable	Not applicable
10 2/	T_A = +120°C; V_{RWM} = 40 V(pk); I_O = 0, half sine wave, f = 60 Hz	T_A = +120°C; V_{RWM} = 40 V(pk); I_O = 0, half sine wave, f = 60 Hz	Not applicable
11	ΔI_{R1} ≤ 100 percent of initial reading or 2 µA, whichever is greater. ΔV_{FM2} ≤ ±50 mV dc.	I_{R1} and V_{FM2}	Not applicable
12	See 4.3.2	See 4.3.2, t = 48 hours	Not applicable
13	Subgroup 2 of table I herein; ΔI_{R1} ≤ 100 percent of initial reading or 2 µA, whichever is greater; V_{FM2} ≤ ±50 mV dc.	Subgroup 2 of table I herein; ΔI_{R1} ≤ 100 percent of initial reading or 2 µA, whichever is greater; V_{FM2} ≤ ±50 mV dc.	Not applicable

1/ Thermal impedance shall be performed any time after sealing provided. Temperature cycling is performed in accordance with MIL-S-19500, screen 3 prior to this thermal impedance.

2/ Junction temperature (T_J) is not to exceed +125°C with V_{RWM} = 40 V(pk). T_J is affected by the device mounting thermal resistance when parasitic power is generated by the temperature dependent leakage current. Until this leakage becomes significant near thermal runaway, T_J remains approximately equal to T_A or T_L for I_O = 0.

4.3.1 Screening (JANHC or JANKC). Screening of die shall be in accordance with MIL-S-19500, appendix H. As a minimum, die shall be 100 percent probed in accordance with group A, subgroup 2.

Source: https://assist.dla.mil -- Downloaded: 202640-19T04:07Z
Check the source to verify that this is the current version before use.

4.3.2 Burn-in conditions. Burn-in conditions are as follows: I_F = 200 mA dc. Mounting and test conditions in accordance with MIL-STD-750, method 1038, test condition B.

4.3.2.1 Mounting. Devices may be mounted using any convenient method including the temporary attachment of leads on US suffix devices, provided that the parts are burned-in at $T_J \geq +110°C$.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. The following test conditions shall be used for $Z_{\theta JX}$, group A inspection:

a. I_M measurement current . . . . . 1 mA to 10 mA.

b. I_H forward heating current . . . . . 0.5 A to 1.0 A.

c. t_H heating time . . . . . 10 ms.

d. t_{MD} measurement delay time . . . . . 70 µs maximum.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
3	4066	I_{FSM} = 5 A(pk), condition A 2, I_O = 200 mA; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); 5 surges of 8.3 ms each at 1 minute intervals.
4	1036	I_O = 200 mA; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); f = 50-60 Hz; V_{RWM} = 40 V(pk); $t_{on} = t_{off}$ 3 minutes minimum for 2,000 cycles.
5	1027	T_A = +100°C ±25°C; I_O = 200 mA (minimum) with I_O adjusted as required to achieve an average lot T_J = +125°C +0°C, -5°C.
6	3101	$R_{\theta JL}$ = 250°C/W; L = 9.52 mm (.375 inch) lead length (non-surface mount). $R_{\theta JEC}$ = 100°C/W (surface mount).

4.4.2.2 Group B inspection, table IVb (JAN, JANTX, and JANTXV of MIL-S-19500).

Subgroup	Method	Condition
2	4066	I_{FSM} = 5 A(pk), condition A 2, I_O = 200 mA; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); 5 surges of 8.3 ms each at 1 minute intervals.
3	1027	I_O = 200 mA; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); f = 50-60 Hz; V_{RWM} = 40 V(pk).
4	2075	In accordance with 4.5.2.

Source: https://assist.dla.mil -- Downloaded: 202610-19T04:07Z
Check the source to verify that this is the current version before use.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Condition
2	2036	Tension: Test condition A; weight = 8 pounds; t = 15 s. Lead fatigue: Test condition E; weight 1 pounds. NOTE: Both tension and lead fatigue are not applicable for US devices.
6	1026	I_O = 200 mA; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); f = 50-60 Hz; V_{RWM} = 40 V(pk).

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with MIL-S-19500 and the conditions for subgroup testing in table II herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in 4.3.2.1 of MIL-STD-750.

4.5.2 Steady-state operation life. This test shall be conducted with a half-sine wave of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall not be greater than 180° nor less than 150°.

4.5.3 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit (not to exceed the group A, subgroup 2 limit) for $Z_{\theta JX}$ in screening (table II of MIL-S-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition.

4.5.3.1 For initial qualification or requalification. Read and record data ($Z_{\theta JX}$) shall be supplied to the qualifying activity on one lot (random sample of 500 devices minimum). Twenty-two serialized devices shall be sent to the qualifying activity for test correlation.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:07Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.5.3	$Z_{\theta JX}$		20	°C/W
Forward voltage	4011	I_{FM} = 20 mA (pk) pulse method (see 4.5.1)	V_{FM1}		0.37	V
	4011	I_{FM} = 200 mA (pk) pulse method (see 4.5.1)	V_{FM2}		0.50	V
	4011	I_{FM} = 630 mA (pk) pulse method (see 4.5.1)	V_{FM3}		0.70	V
Reverse current leakage	4016	V_{RM} = 40 V (pk) pulse method (see 4.5.1)	I_{RM1}		5.0	µA
Subgroup 3						
High temperature operation:		T_A = +100°C				
Reverse current leakage	4016	V_{RM} = 40 V (pk) pulse method (see 4.5.1)	I_{RM2}		0.60	mA
Forward voltage	4011	I_F = 200 mA (pk) pulse method (see 4.5.1)	V_{FM4}		0.47	V
Low temperature operation:		T_A = -55°C				
Reverse current leakage	4016	V_{RM} = 40 V (pk) pulse method (see 4.5.1)	I_{RM3}		20	µA
Forward voltage	4011	I_F = 200 mA (pk) pulse method (see 4.5.1)	V_{FM5}		0.60	V
Subgroup 4						
Capacitance	4001	V_R = 0 V dc .01 ≤ f ≤ 1 MHz V_{SIG} = 15 mV p-p	C_T		50	pF

1/ For sampling plan, see MIL-S-19500.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:07Z
Check the source to verify that this is the current version before use.

TABLE II. Group E inspection (all quality levels) for qualification only. 1/

Inspection 2/	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices c = 0
Thermal shock (temperature cycling)	1051	500 cycles	
Hermetic seal	1071	Test condition E	
Electrical measurement		See table I, group A, subgroup 2	
Subgroup 2			22 devices c = 0
Steady-state reverse bias	1038	test condition A; 1,000 hours, See 4.3, screen 10	
Electrical measurement		See table I, group A, subgroup 2	
Subgroup 3			
Not applicable			
Subgroup 4			22 devices c = 0
Thermal resistance, (forward voltage drop diode method)	4081 or 3101	$R_{\theta JL}$ = 250°C/W maximum at 9.52 mm (.375 inch) lead length; $R_{\theta JEC}$ = 100°C/W maximum; method 3101 in accordance with 4.5.3 except I_H = 200 mA and T_H = 20 s (minimum)	c = 0
Subgroup 5			
Not applicable			
Subgroup 6			
Not applicable			

1/ For initial design and process change verification only (one time testing).
2/ For sampling plan, see MIL-S-19500.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:07Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Product assurance level; type designator; and for die acquisition, the JANHC and JANKC identification (see 6.3 and figures 3 and 4).

6.3 Suppliers of die. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N6677) will be identified on the QPL.

JANC ordering information		
	Manufacturer	
PIN	55801	---
1N6677-1	JANHCA1N6677-1	---
	JANKCA1N6677-1	---

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 85, 99

Preparing activity:
DLA - ES

(Project 5961-1679)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:07Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/6108	2. DOCUMENT DATE (YYMMDD) 94/07/22

3. DOCUMENT TITLE

SEMICONDUCTOR DEVICE, HERMETIC, DIODE, SILICON, SCHOTTKY BARRIER, TYPES 1N6677-1 AND 1N6677UR-1 JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial: 513-296-6048 (2) AUTOVON: 986-6048
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center Attn: DESC-ELDT Dayton, Ohio 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:07Z
Check the source to verify that this is the current version before use.